

Ascent trajectory design and optimization of a two-stage throttleable liquid rocket

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Abstract

This paper explores in detail the design and optimization of the ascent trajectory for a two-stage liquid rocket capable of providing variable thrust. Thrust and steering profiles play an essential role in designing a launch vehicle mission. Variable thrusting or throttling requirement arises in the thrust profile of a liquid rocket due to structural, aerodynamic and mission-related constraints. The current study investigates this aspect for two-stage liquid rockets, which deliver a 500 kg payload to 500 km circular orbit and 180×36000 km Geostationary Transfer Orbit (GTO). As payload mass is kept constant in both cases, the lift-off mass varies. The study found that for both missions mentioned earlier, the required variation of thrust (throttling) is in the range of 100% to 33% for the first stage to achieve an optimal trajectory. For the second stage, a considerable requirement for throttling is absent. Optimal trajectory with throttling capability reduced the lift-off mass of the launch vehicle for a mission to circular orbit by 41.6% and to the GTO by 16.3% compared to the constant thrusting cases. Also, a two-stage liquid rocket using semi cryogenic engines with variable thrusting capability and stage structural factors as low as 0.1 can replace a three-stage rocket which delivers the same payload to the 500 km circular orbit. Kulasekharapattinam and Sriharikota were the launch sites considered for the circular orbit and GTO. The study did not consider spent stage impact constraints. Canonical Differential Evolution based solver is used for global optimization. Thrust profile, steering profile and lift-off mass are the outputs of the optimization strategy.

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1. Introduction

The objective of the current study is to design and optimize the thrust profile and trajectory of a two-stage liquid propelled launch vehicle. In the typical operational scenarios, optimization is performed only for the trajectory, considering Euler angle/body rates as variables where the thrust profile is predefined. This study analyses a variable-thrust profile and trajectory requirement, which would yield a higher payload fraction. The thrust profile

is considered a variable in the optimization process. The throttling capability of the liquid rocket's first or booster stage and/or upper stage(s) is an essential requirement if variable thrusting has to become operational. Varying thrust for the Ascent Trajectory of a rocket with the launch vehicle and mission constraints is a complex non-linear problem.

Solid rocket motor powered stages can provide a varying thrust profile through the grain design. [Hartfield et al. \(2004\)](#) have developed an analytical model for the design of slotted-grain solid rocket motors for the desired thrust profile. Most liquid rocket engines are designed to operate at a specified chamber pressure for optimum thrust

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Nomenclature

m	Mass	$\mathbf{x}$	Target vector/ Design variables
c	Velocity of sound	$\mathbf{v}$	Mutant vector
h	Altitude	$\mathbf{u}$	Trial vector
α	Angle of attack	G	Generation
β	Sideslip angle	D	Dimension
γ	Flight path angle or Elevation angle	$f(x)$	Objective function
λ	Azimuth angle	E, P	No. of end constraints, No. of path constraints
ϕ	Roll angle	<i>Subscripts</i>	
θ	Pitch angle	I	Inertial or Earth centered inertial frame
ψ	Yaw angle	R	Relative
σ	Bank angle	A	Aerodynamic or Atmospheric relative frame
ϕ_c	Geocentric latitude	L	Launch frame
ϕ_g	Geodetic latitude	B	Body frame
θ_I	Inertial longitude	D	Drag
I_{sp}	Specific impulse	S	Steering
s_f	Structural factor	G	Gravity
N_S	No. of stages	$1, 2, \dots, pn$	Pitchrate segment in the n^{th} stage
ζ_{PL}	Payload fraction	$1, 2, \dots, Tn$	Thrust segment in the n^{th} stage
$\mathbf{T}$	Thrust vector	i, j	Index of parameter in a parent
$\mathbf{D}$	Drag vector	x	Target vector
$\mathbf{a}$	Acceleration vector	R	Index of parent in Target vector
$\mathbf{v}$	Velocity vector	LO	Lift-Off
$\mathbf{r}$	Position vector	min	Minimum
u, v, w	x, y, z components of velocity vector	max	Maximum
ΔV	Added velocity	u	Values
T/F	Thrust	e	Index of equality constraint
D	Drag	p	Index of inequality constraint, Perigee
$\mathbf{g}$	Acceleration due to gravity vector	a	Apogee
V	Velocity	<i>Superscripts</i>	
t	Time	G	Generation
$\dot{\theta}$	Pitchrate	f	final
DE	Differential evolution		
GA	Genetic algorithm		
$\mathbf{e}$	Allowed deviation in terminal constraints		

without throttling. Combustion instability, chamber pressure roughness, flow separation and stall of the pump (in the case of pump-fed engines) are the challenges liquid engines face to achieve throttled operation by varying the chamber pressure. In the past, both pump-fed and pressure-fed engines have demonstrated deep throttling capability in a range up to 10:1. [Giuliano et al. \(2010\)](#) mentions that the Common Extensible Cryogenic Engine or the CECE, which is a modified version of the expander-cycle and pump-fed RL-10 engine of the Atlas-V launch vehicle, is capable of deep throttling. [Dressler \(2006\)](#) explains the 10:1 deep throttling capability of the Lunar Module Descent Engine (LMDE), which operates using a pressure-fed cycle. Apollo moon landing mission used the TRW designed LMDE, which employed a single element coaxial pintle injector element in conjunction with variable area cavitating venturi flow control valves. [Bjelde et al. \(2008\)](#)

mentions that the SpaceX Falcon 9 Merlin engines also utilize a pintle injector which has achieved a throttling up to 60%.

The Atlas/Centaur vehicle used the Augmented Lagrangian method for trajectory optimization with a predefined thrust profile. The solution was obtained by alternately minimizing the augmented function for the independent variable and adjusting the Lagrange multipliers to satisfy the constraints. NASA's POST (Program to Optimize Simulated Trajectories) by [Brauer et al. \(1975\)](#) employs an Accelerated Projected Gradient method for optimization. The methods above are linear optimization tools that take considerable computation time for convergence. Quadratic programming also has been attempted in the problem, considering the low computational cost. ISRO's PYOPT (Pitch and Yaw steering program Optimization), which is a trajectory optimization program with pre-defined thrust profile,

uses Diagonalized Multiplier Methods (DMM) as mentioned by [Adimurthy et al. \(2013\)](#). [Adimurthy \(1976\)](#) mentions the optimization of a launch vehicle ascent trajectory using Euler angle rates as the input variables. [Zheng et al. \(2020\)](#) used hierarchical optimization, which is a numerical method. [Chai et al. \(2018a\)](#) applied a similar modelling technique as explained in this paper by dividing the flight sequence into discrete phases. The problem was constructed as an optimal control formulation. A newly proposed gradient-based optimization algorithm was used to solve the optimization problem. Heuristic algorithms also have been applied to the trajectory optimization problems. [Zardashti and Rahimi \(2020\)](#) have used network flow-based algorithm for ascent trajectory optimization. The disadvantages are the increase in complexity due to the size of the co-state matrix, sensitivity to the initial guess and possibility of convergence to local minima. [Chai et al. \(2019\)](#) mentions the application of stochastic algorithms for trajectory optimization problems. Genetic Algorithm (GA), Differential Evolution (DE), Violation Learning Differential Evolution (VLDE), Particle Swarm Optimization (PSO), Ant Colony (AC) Optimization, Artificial Bee Colony (ABC) optimization are to name a few. [Chai et al. \(2019\)](#) highlights that, compared with classic gradient-based algorithms, as the initialization of the population is random, there is no initial guess value required by stochastic algorithms. Also, they are more likely than gradient-based algorithms to locate the global minimum. [Federici et al. \(2021\)](#) utilized DE for global optimization of Solid Rocket Motor (SRM) grain design and trajectory of a Vega-derived light launch vehicle. [Ghosh and Chattopadhyay \(2014\)](#) has proposed a Differential Evolution based Trajectory Optimisation of a four-stage launch vehicle. It also compares optimization based on Genetic Algorithm and Differential Evolution. Constraints were handled using penalty methods. The objective function included injection altitude, injection velocity, injection angle and the required orbital velocity as the variables. The absolute difference between required and obtained variables' values was added using weighting factors to obtain the objective function. The study neither attempted to minimize the vehicle mass nor optimize the thrust profile. However, the study established that DE is a more efficient optimization method when compared to GA in terms of speed of convergence. [Das et al. \(2016\)](#) mentions DE as the most powerful and versatile evolutionary algorithm for continuous parameter spaces. A newly formulated Evolutionary Optimization at Sapienza (EOS), as mentioned by [Federici et al. \(2020\)](#) which is an improvement to Differential Evolution, was utilized in a highly constrained space trajectory optimization problem. [Chai et al. \(2017a\)](#) utilized Violation Learning Differential Evolution (VLDE) to generate an initial guess for solving a space maneuver vehicle's trajectory optimization using direct numerical method. The paper presented a combinatory approach that uses the VLDE algorithm and the hp-adaptive pseudospectral method. The methodology adopted a modified evolu-

tionary strategy, and the amount of constraint and fitness violations were quantified to get a better convergence speed. The evolutionary strategy included selection, direct search using the Nelder-Mead algorithm and adaptive mutation. [Chai et al. \(2018b\)](#) describes a technique for multi-objective optimization for a vehicle's trajectory. A fuzzy goal programming-based (FGP) gradient hybrid genetic algorithm was applied. [Chai et al. \(2017b\)](#) used a modified game theory approach coupled together with an adaptive differential evolution algorithm to generate a Pareto front for a multi-objective trajectory optimization problem. [Ponssard et al. \(2009\)](#) studied the effect of variable thrusting for the Ariane 5 launcher for a GTO mission. Although Ariane 5 rocket is a two-stage liquid rocket and the mission requirement is similar to the case analyzed in our study, the launch site, constraints, design variables, and optimization strategy were different from the study presented in this paper. The study did not apply path constraints, which directly impacted the results. More details on this aspect are discussed further in the paper.

Differential Evolution was introduced as a new heuristic approach for minimizing non-linear and non-differentiable problems in 1995 by [Storn and Price \(1997\)](#). The method is well suited for cluster computing as the algorithm supports parallel evaluations. The leading search engine could be written in less than 30 lines. The property of vector computation in DE could iterate and generate solutions in shallow regions of fitness. Objective values diverge by increasing standard deviation in shallow areas or local minima, finally leading to a global optimum. Differential Evolution (DE) provides flexibility to the offspring generation strategy due to the variability in the Mutation and Crossover ratios. Different combinations of target vectors spread throughout the design space through differential mutation make the program arrive at the global optimum. The crossover ratio enables the user to decide the portion of information inherited by the trial vector from the target vector. The result search moves are intuitive. It makes DE a versatile and robust optimization strategy for a wide range of complex optimization scenarios as explained by [Das et al. \(2016\)](#). Differential Evolution is one among the top three optimization algorithms in the CEC competitions since 2005, except in 2013 where it came 4th as shown by [Pant et al. \(2020\)](#). [Chai et al. \(2021\)](#) mentioned that evolutionary algorithms are more likely to locate global minimum for space/aerospace missions. [Chai et al. \(2019\)](#) explains that although heuristic-based optimization methods are feasible for addressing trajectory design problems, the validation of solution optimality becomes difficult. Even though the algorithm is simple, the computational cost and complexity tend to be high.

In the current paper, DE/rand/1/bin strategy as mentioned in [Storn and Price \(1997\)](#) is utilized in the solution. It's also referred as Canonical DE in [Poole and Allen \(2019\)](#). The method follows a 5 point algorithm involving the following:

1. Initialization
2. Mutation
3. Crossover
4. Selection
5. Stopping

The study in this paper did not consider structural forces, aero heating and spent stage impact constraints.

A 3 DOF multi-stage launch vehicle simulator was developed as part of the current study, which was inspired by the Program to Optimize Simulated Trajectories (POST, Volume 1) by Brauer et al. (1975). The main program can provide instantaneous vehicle attitudes, rates and coordinates for varying thrust and Euler angle inputs. A staging optimization subroutine based on sequential quadratic programming was developed to divide the propellant mass and structural mass among the stages with different structural factors and specific impulses. The main optimizer subroutine constantly interacts with the vehicle simulator and compares the output with the objective requirement and multiple non-linear and non-differentiable constraints, making the program capable of multi-constraint global optimization. Terminal/End constraints were applied in altitude, injection angle, injection velocity, apogee and perigee altitudes. Path constraints were applied in peak acceleration, peak dynamic pressure and angle of attack.

Compared to the optimization technique used in Chai et al. (2017a) which has improved population diversity and convergence speed, the method used here is the classical method mentioned in Storn and Price (1997). It is the most straightforward DE algorithm. Both methods did not employ penalty based constraints which avoided the design of penalty factors. Here, the focus of the study was on assessing the advantage of an improved trajectory design using throttling, which needed a simple and well-established optimization algorithm. Due to the increased number of variables while simultaneously optimizing thrust profile and trajectory, gradient-based methods were not found suitable for the optimization approach, as calculating the rates of each variable would make the logic cumbersome. Hence, the study in the current paper did not consider hybrid and combinatorial evolutionary algorithms and classical optimization methods. Since the objective of the current research was to carry out a single objective optimization for a problem with multiple variables and constraints, and the main focus of the study is to apply a standard Differential Evolution technique to bring out the optimum configuration for a liquid rocket employing variable thrust during its operation, the simplest design of Differential Evolution algorithm was applied in the optimization process.

2. Simulator design

The launch vehicle simulator simulates 3-dimensional point mass trajectories in powered flight. The earth is considered a rotating oblate spheroid by the simulator. The program uses a planet model, gravitational model and

atmospheric model as mentioned in Brauer et al. (1975). The oblate spheroid model computes the oblate planet geometry, and the atmospheric model computes the ambient pressure, density, temperature and speed of sound. The gravitational model computes the gravitational acceleration as the altitude varies. The launch was divided into 4 phases:

1. Vertical ascent.
2. Kickrate maneuver.
3. Gravity Turn
4. Optimal Trajectory

The vertical ascent phase is used by the launch vehicle to clear the launch pad. The Euler angles before and after the vertical ascent remains the same. Kickrate maneuver is implemented to impart an initial turn to the vehicle w.r.t to the local vertical to initiate gravity turn. Kickrate is the rate at which the vehicle pitches during the maneuver. Gravity turn is a zero-lift turn maneuver which utilizes the earth's gravitational force to tilt the vehicle. During this phase, the vehicle's angle of attack remains at zero. The vehicle uses gravity turn during the atmospheric phase to reduce the aerodynamic forces acting on it. Optimal trajectory is the final phase of the launch vehicle mission. It makes sure that the payload reaches the required terminal conditions with minimum energy utilization.

The simulator uses five coordinate systems. Fig. 1 shows the Earth Centred Inertial (I) and Launch Pad Inertial (L) reference frames. Fig. 2 shows the Body Centred, Topocentric Horizontal or Geographic and Aerodynamic Frames which are moving reference frames. Fig. 3 shows the moving reference frames and the Earth Centred Inertial frame. Gravitational forces are calculated in the I frame. The body forces resulting from thrust and drag are calculated in the Body Centred frame. The resultant

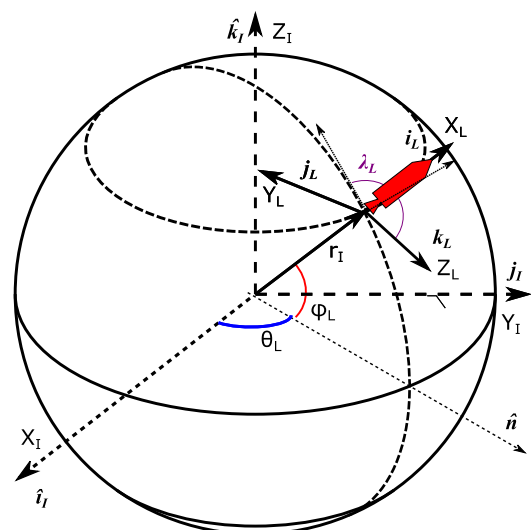


Fig. 1. Earth Centred Inertial (I) & Launch Pad Inertial (L) frames.

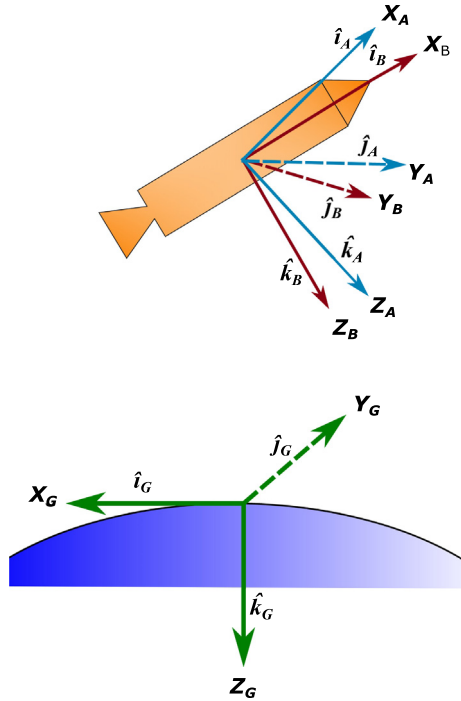


Fig. 2. Body Centered (B), Aerodynamic (A) and Geographic (G) frames.

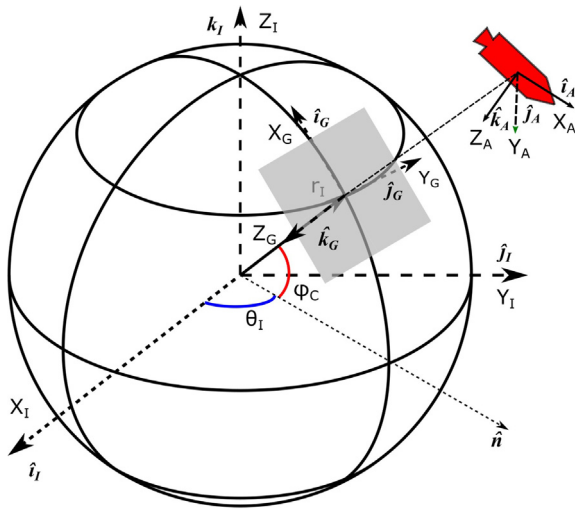


Fig. 3. Earth Centered Inertial (I), Geographic (G) & Aerodynamic (A) frames.

force on the launch vehicle body is obtained by transforming the Body forces from the Body Centered frame to Earth Centered Inertial frame. The translational equations of motion are solved in this frame. Euler angles are estimated from the rotation of Body Centred Frame w.r.t the L frame. Aerodynamic angles are estimated from the rotation of Body Centred Frame w.r.t the Aerodynamic frame. ϕ_c and θ_I are used to estimate the instantaneous attitude of the Geographic frame. γ and λ are estimated from the Geographic frame.

The simulation does not include coasting and stage separation times. The fundamental character of the design lies

in the division of the trajectory into a sequence of simulation segments. By using segments, each phase is modelled suitably according to operational requirements. Right Hand Coordinate System (RHS) is used to define the coordinate frame axes. Defined steering rates or angles are used to maneuver the vehicle. The types of steering options available in the simulator are:

1. Inertial Euler angles/rates based steering.
2. Aerodynamic angles/rates based steering,

Vertical Ascent, Kickrate Maneuver and Optimal Trajectory follow the same set of equations steered based on Inertial Euler angles and rates. Gravity turn is steered based on Aerodynamic angles and rates.

The steering option using inertial Euler angles provides a straightforward approach as the transformation from Earth Centered Inertial to Body Centred frame ([IB]) is based on the input angles/rates. While steering using aerodynamic angles, a series of transformations are required to obtain [IB], which is critical to the calculations. However, aerodynamic and inertial Euler angles based steering are required to design the launch vehicle trajectory. The flow charts of both types of steering are shown in Figs. 4 and 5. As seen in Fig. 4, the final Euler angles are obtained by integrating the initial inertial Euler angles and rates, which calculate [IB]. [IB] is utilized to convert the body forces to the ECI frame. Fig. 5 shows the implementation of aerodynamic angles based steering. The final aerodynamic angles are obtained by integrating the initial aerodynamic angles and rates, which calculates the [AB] transformation matrix. [GB] is computed from the initial elevation and azimuth angles and [AB]. Eqs. (1)–(24) show the calculations.

2.1. Equations of motion

$$\mathbf{D}_{AB} = c_D \times A \times q_\infty (1)$$

$$\mathbf{a}_B = [\mathbf{F}_{TB} - \mathbf{D}_{AB}] / \dot{m} (2)$$

$$\mathbf{a}_I = [\mathbf{IB}]^T \times \mathbf{a}_B + \mathbf{g} (3)$$

$$\mathbf{v}_I = \int \mathbf{a}_I dt + \mathbf{v}_{I,0} (4)$$

$$\mathbf{r}_I = \int \mathbf{v}_I dt + \mathbf{r}_{I,0} (5)$$

2.2. Inertial Euler angles/rates based steering

$$\phi_{i+1} = \phi_i + \dot{\phi} \cdot dt (6)$$

$$\theta_{i+1} = \theta_i + \dot{\theta} \cdot dt (7)$$

$$\psi_{i+1} = \psi_i + \dot{\psi} \cdot dt (8)$$

$$\theta = \alpha + \gamma (9)$$

2.3. Aerodynamic angles/rates based steering

$$\alpha = \alpha_i \quad (10)$$

$$\beta = \beta_i \quad (11)$$

$$\sigma = \sigma_i \quad (12)$$

$$[GB] = [AB] \times [GA] \quad (13)$$

$$\mathbf{v}_{A,B} = [GB]\mathbf{v}_{A,G} \quad (14)$$

$$\mathbf{v}_{A,L} = [IL]\mathbf{v}_{A,I} \quad (15)$$

$$\hat{v}_{A,L} = \frac{\mathbf{v}_{A,L}}{\|\mathbf{v}_{A,L}\|} \quad (16)$$

$$\hat{v}_{A,B} = \frac{\mathbf{v}_{A,B}}{\|\mathbf{v}_{A,B}\|} \quad (17)$$

$$s = \hat{v}_{A,B} \times \hat{v}_{A,L} \quad (18)$$

$$c = \hat{v}_{A,B} \cdot \hat{v}_{A,L} \quad (19)$$

$$[K] = \begin{bmatrix} 0 & -s_3 & s_2 \\ s_3 & 0 & -s_1 \\ -s_2 & s_1 & 0 \end{bmatrix} \quad (20)$$

$$[LB] = \frac{K^2}{(1+c)} + K + I_3 \quad (21)$$

$$\phi = \tan^{-1} \left[\frac{LB_{23}}{LB_{22}} \right] \quad (22)$$

$$\theta = \tan^{-1} \left[\frac{LB_{31}}{LB_{11}} \right] \quad (23)$$

$$\psi = \sin^{-1} (LB_{21}) \quad (24)$$

The calculation of aerodynamic angles, flight path or elevation and azimuth angles, geocentric latitude and inertial longitude and coordinate transformations were done as explained in Brauer et al. (1975). However, quaternion based equations were not implemented.

3. Basic formulation of the optimization problem

The design was implemented as a single objective optimization problem with an optimizer using Canonical Differential Evolution. The problem can be expressed as follows:

Find design variables,

$$\mathbf{x} = (x_1, x_2, \dots, x_D) \quad (25)$$

To minimize Objective Function,

$$f(x) = m_{LO} \quad (26)$$

Subjected to,

$$\mathbf{x}_{\min} \leq \mathbf{x} \leq \mathbf{x}_{\max} \quad (27)$$

$$h_e(x) = 0 \quad e = 1, 2, \dots, E \quad (28)$$

$$g_p(x) \leq 0 \quad p = 1, 2, \dots, P \quad (29)$$

The design variables are included in the Target Vector $\mathbf{x}$. The objective function is the rocket's lift-off mass. The variables in the target vector are allowed to vary within the design space $[\mathbf{x}_{\min}, \mathbf{x}_{\max}]$. The global optimization algorithm

varies the design variables to minimize the objective function subject to the terminal constraints $h(\mathbf{x})$ and path constraints $g(\mathbf{x})$.

4. Global optimizer algorithm

$$\mathbf{x} = \left\{ \begin{array}{c} \Delta v_{ideal} \\ \dot{\theta}_1^k \\ \dot{\theta}_1^{s1}, \dot{\theta}_2^{s1}, \dots, \dot{\theta}_{p1}^{s1} \\ \dot{\theta}_1^{s2}, \dot{\theta}_2^{s2}, \dots, \dot{\theta}_{p2}^{s2} \\ \dots \dots \dots \\ T_1^{s1}, T_2^{s1}, \dots, T_{T1}^{s1} \\ T_1^{s2}, T_2^{s2}, \dots, T_{T2}^{s2} \\ \dots \dots \dots \\ \Delta t_{\theta,1}^k \\ \Delta t_{\theta,1}^{s1}, \Delta t_{\theta,2}^{s1}, \dots, \Delta t_{\theta,p1}^{s1} \\ \Delta t_{\theta,1}^{s2}, \Delta t_{\theta,2}^{s2}, \dots, \Delta t_{\theta,p2}^{s2} \\ \dots \dots \dots \\ \Delta t_{T,1}^{s1}, \Delta t_{T,2}^{s1}, \dots, \Delta t_{T,T1}^{s1} \\ \Delta t_{T,1}^{s2}, \Delta t_{T,2}^{s2}, \dots, \Delta t_{T,T2}^{s2} \\ \dots \dots \dots \end{array} \right\} \quad (30)$$

Eq. (30) shows the definition of the target vector for the design and simultaneous optimization of thrust profile and trajectory. Kickrate, pitch rates and thrusts are segmented, and their magnitudes and durations are used as design variables in the target vector. End constraints and path constraints were applied to the problem as shown in Table 5. The operations performed on the target vector to achieve optimal solution are discussed further.

4.1. Initialization

Initialization is the generation of random distribution of N_p real-valued target vectors of D dimension. Each target vector is a candidate for the solution. Each candidate solution has a length of D . Initialization is denoted by $G = 0$ generation. The generation of random distribution is uniform unless otherwise stated. The design space is denoted by Ω , which can be,

$$\Omega \subseteq R^D \quad (31)$$

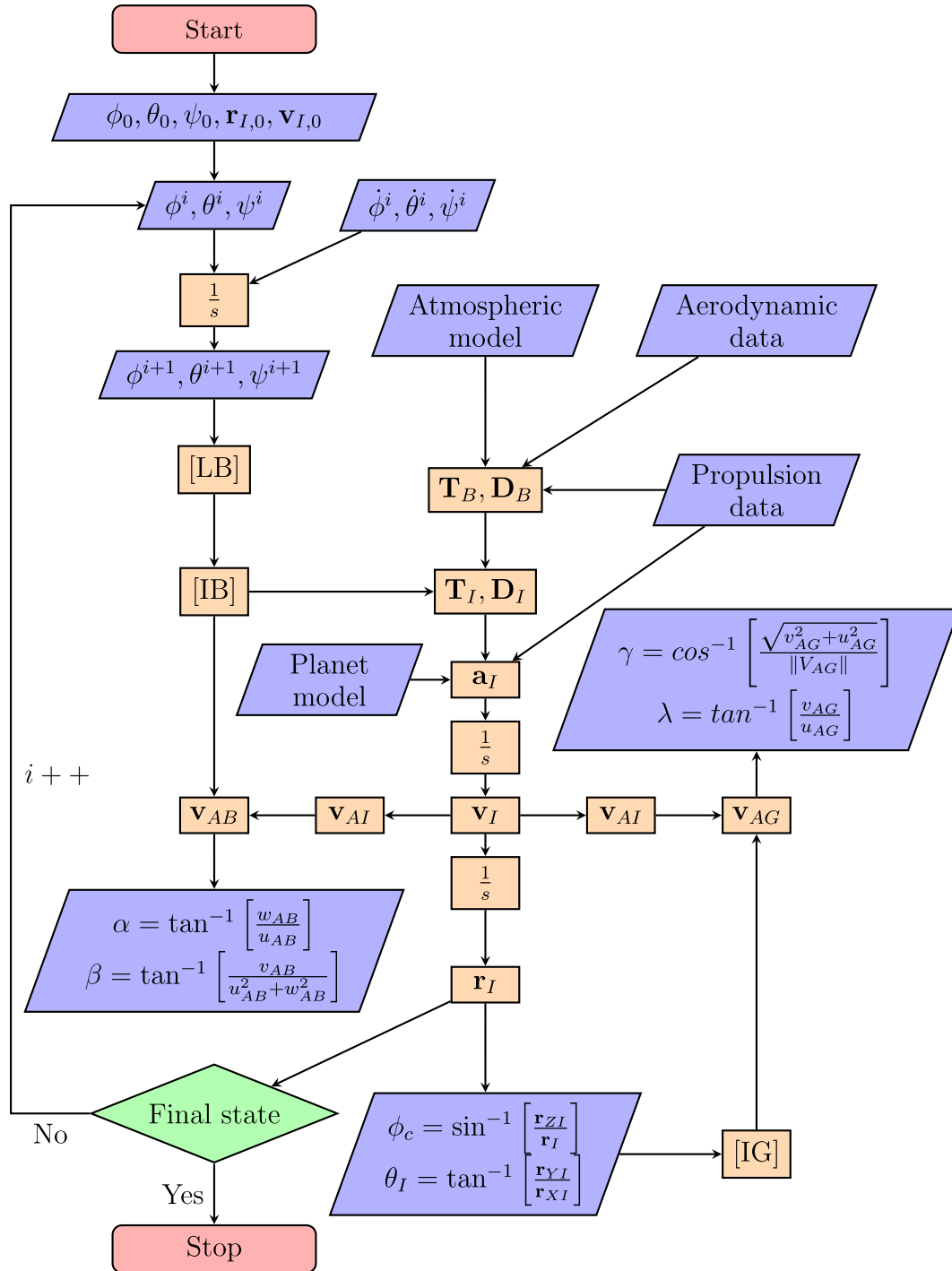


Fig. 4. Flow chart for Inertial Euler angles/rates based steering.

The family of target vectors are denoted by:

$$\mathbf{x}^G = (x_1^G, x_2^G, \dots, x_{N_p}^G) \quad (32)$$

where G is the number of iteration/generation denoted by:

$$G = 0, 1, 2, \dots, G_{max} \quad (33)$$

Initialization is done such that the randomly selected candidates span the entire design space. For initialized target vector is:

$$\mathbf{x}^0 = (x_1^0, x_2^0, \dots, x_{N_p}^0) \quad (34)$$

After initialization, the objective function for the Target Vector s is evaluated. After evaluation, the objective values of the Target Vector s are denoted as:

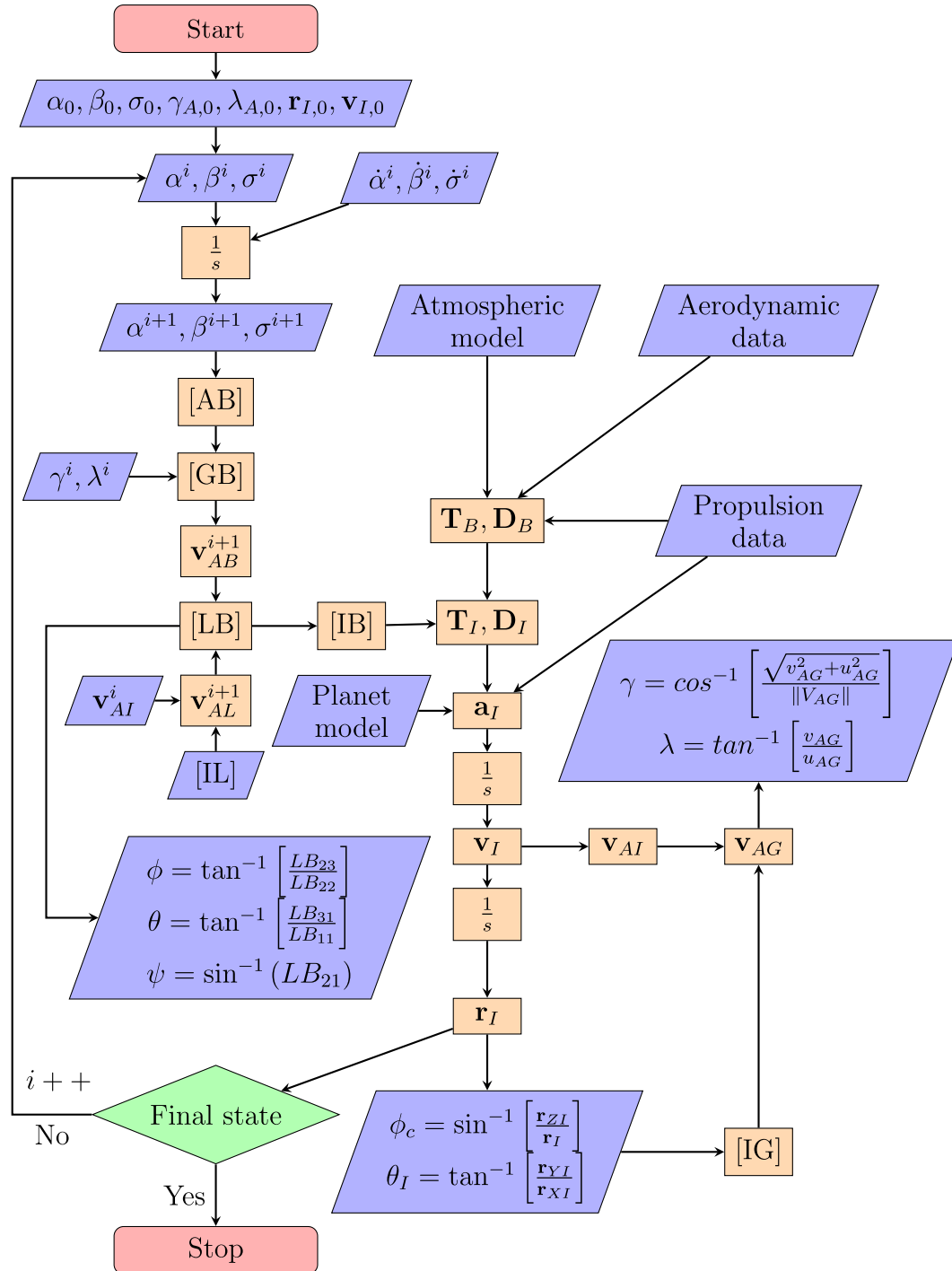


Fig. 5. Flow chart for Aerodynamic angles/rates based steering.

$$f_x^G = (f_{x,1}^G, f_{x,2}^G, \dots, f_{x,N_p}^G) \quad (35)$$

Where, each term in f_x^G is a scalar. The constraint values are denoted as,

$$c_x^G = (c_{x,1}^G, c_{x,2}^G, \dots, c_{x,N_p}^G) \quad (36)$$

Where, each term in c_x^G is a non-dimensionalized sum of individual constraint violations. Each candidate solution has the number of constraints as mentioned in Eqs. (28) and (29).

Once the initialization is completed, the optimization moves to recombination to generate a Trial Vector for each Target Vector. Initialization is only done once, whereas recombination repeats till the optimization finishes. Muta-

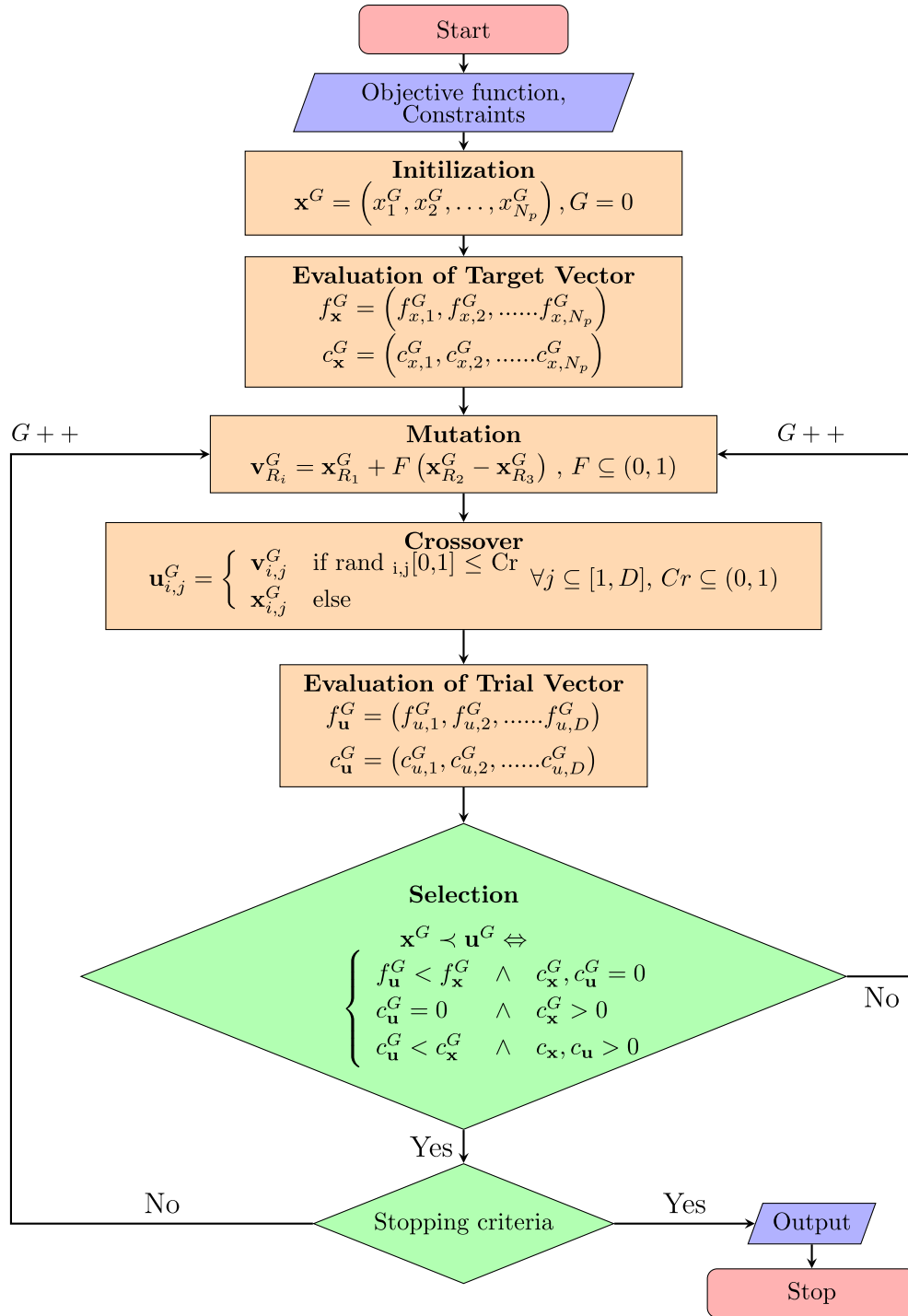


Fig. 6. Flow chart for Optimization with Differential Evolution.

tion and Crossover are the steps in recombination. The sections below explain the details.

4.2. Mutation

Mutation creates a Mutant Vector $\mathbf{v}^G$ corresponding to each $\mathbf{x}^G$ in the current generation. Canonical DE technique is as follows:

$$\mathbf{v}_{R_i}^G = \mathbf{x}_{R_1}^G + F(\mathbf{x}_{R_2}^G - \mathbf{x}_{R_3}^G) \quad (37)$$

The name 'Differential Evolution' comes from the differential mutation strategy as stated above. The indices R_1, R_2, R_3 are mutually exclusive integers randomly chosen from range $[1, N_p]$, and all are different from the base index i and are made randomly for each Target Vector. The

Table 4
Vehicle definition.

Case	Orbit, km	N _S	I _{sp} , s	s _f	λ _L [°]	φ _{C,L} [°]	θ _{f,L} [°]	Launchsite
I	500×500	1	270	0.15	180	8.40	78.05	Kulasekarapattinam
		2	310	0.15				
		3	310	0.15				
II	500×500	1	310	0.10	180	8.40	78.05	Kulasekarapattinam
		2	330	0.13				
III	180×36000	1	310	0.10	104	13.73	18.23	Sriharikotta
		2	330	0.13				

Table 5
Mission constraints.

Case	h _i (x)		g _i (x)	
I	h, kms	500±10	a _{max} , g	<6
	γ, °	0±2		
	V _I , m/s	7612±30		
II	h, kms	180±10	a _{max} , g	<6
	γ, °	0±2	Q _{max} , kPa	<100
	V _I , m/s	30		
III	h _a , kms	36000±1000	a _{max} , g	<8
	h _p , Kms	180±10	Q _{max} , kPa	<100
			α (alt. < 110 kms), °	<10

resulting vector is called a recombinant mutant vector. Mutation Factor F is a real and constant value denoted as:

$$F \subseteq (0, 1) \quad (38)$$

is generated randomly for each i in $\mathbf{x}_i^G$. F is very critical in obtaining the solution of optimization. A high value of F amplifies the difference and makes the mutation $\mathbf{x}_{R_2}^G - \mathbf{x}_{R_3}^G$ explorative, whereas a low value of F makes the mutation excavative.

4.3. Crossover

Crossover mixes the components of the Mutant Vector with that of the Target Vector to form a Trial Vector. It is done to increase the diversity in the Target Vector. Binomial Crossover is used based on a fixed parameter called Crossover Ratio (Cr). Cr decides the portion of parameters passed onto the Trial Vector from the Target Vector. The scheme is represented as:

$$\mathbf{u}_{i,j}^G = \begin{cases} \mathbf{v}_{i,j}^G & \text{if rand}_{i,j}[0, 1] \leq \text{Cr} \\ \mathbf{x}_{i,j}^G & \text{else} \end{cases} \quad \forall j \subseteq [1, D] \quad (39)$$

Table 6
Segmentation for Case II.

No. of segments	Stage 1	Stage 2
Kickrate	1	0
Pitchrate	5	5
Thrust	5	5

Table 7
Segmentation for Case III.

No. of segments	Stage 1	Stage 2
Kickrate	1	0
Pitchrate	2	4
Thrust	4	4

Table 8
Optimization parameters.

Case No.	Orbit requirement	N _S	F	Cr	D	N _P
I	500×500 Circular	3	0.4	0.9	35	127
II	500×500 Circular	2	0.4	0.9	43	128
III	180×36000 GTO	2	0.4	0.9	22	120

Table 9
Lift-off Mass comparison for constant thrust and variable thrust cases.

Mission	Constant thrust	Variable thrust
Case II	98850 kgs	57673 kgs
Case III	186063 kgs	155664 kgs

$$\text{Cr} \subseteq (0, 1) \quad (40)$$

4.4. Selection

After recombination as mentioned in *Mutation* and *Crossover*, the objective values (f) and constraint values (c) for the Trial Vector are represented as follows:

$$f_u^G = (f_{u,1}^G, f_{u,2}^G, \dots, f_{u,D}^G) \quad (41)$$

$$c_u^G = (c_{u,1}^G, c_{u,2}^G, \dots, c_{u,D}^G) \quad (42)$$

That is followed by choosing the most appropriate member ($\square_i$) between the Target Vector s and Trial Vector s . Feasibility criteria as explained in Eq. 43 are applied in the selection process.

$$\mathbf{x}^G \prec \mathbf{u}^G \iff \begin{cases} f_u^G < f_x^G & \wedge & c_x^G, c_u^G = 0 \\ c_u^G = 0 & \wedge & c_x^G > 0 \\ c_u^G < c_x^G & \wedge & c_x, c_u > 0 \end{cases} \quad (43)$$

Table 10
Results of optimization with variable thrust.

N_s	Orbit (kms)	Generations	σ_f	m_{PL} , kgs	m , kgs	ζ_{PL}
3	500×500	2.1×10^6	0.0020	585	55398	0.0106
2	500×500	3.15×10^6	0.0001	585	57673	0.0101
2	180×36000	8×10^3	0.0005	585	155664	0.0037

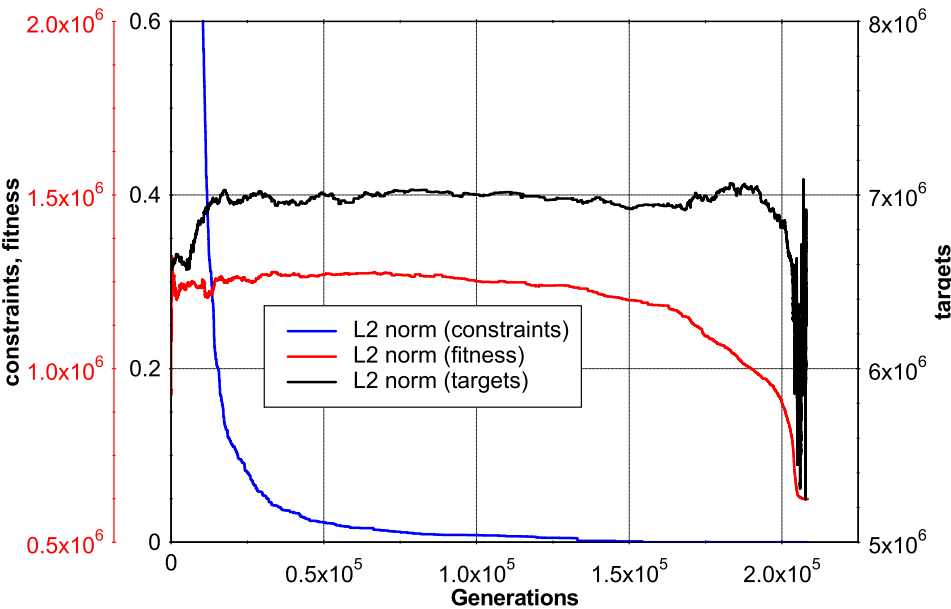


Fig. 7. DE Optimization result for Case I.

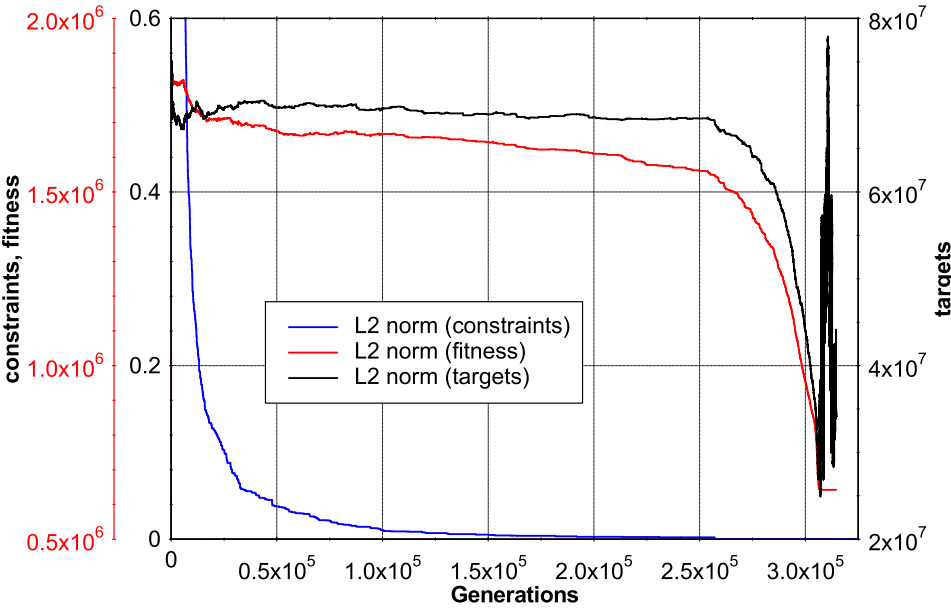


Fig. 8. DE Optimization result for Case II.

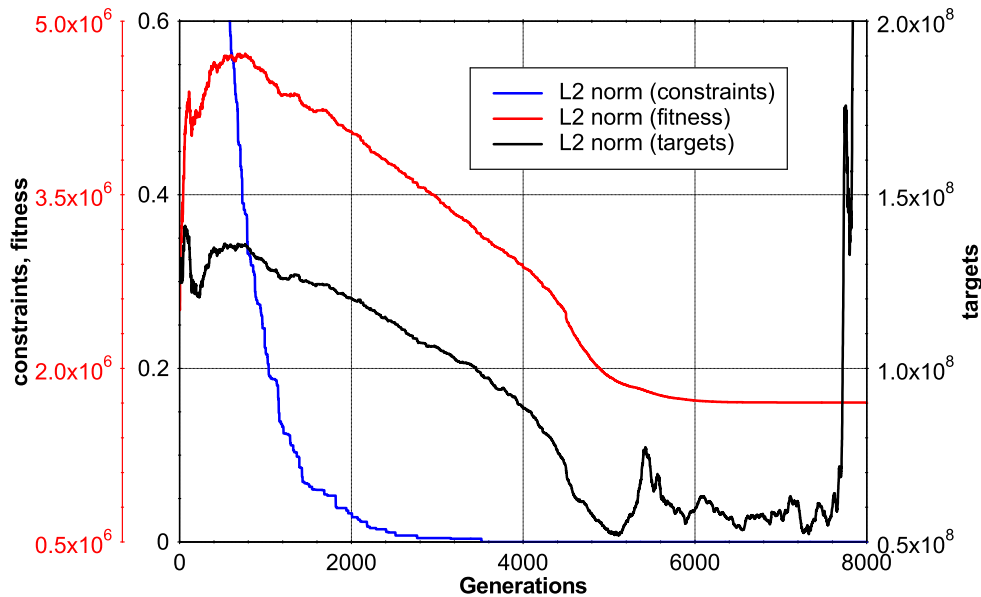


Fig. 9. DE Optimization result for Case III.

Once the selection process is finished, the optimization marches into the next generation. The steps mentioned in sections *Mutation*, *Crossover* and *Selection* are repeated till the optimization is complete. The process of optimization with Differential Evolution is shown in Fig. 6.

4.5. Stopping criteria

Objective value (f) based stopping criteria are applied. Standard deviation (σ_f) of f of each generation is calculated. Optimization is stopped when (σ_f) is lower than the threshold value of 10^{-3} . Another method uses the standard deviation bound for the targets using the same logic. It could be a more precise approach as similar objective values could have different target vectors. It is suitable for both simple and complex problems.

5. Results and discussion

Differential Evolution based Optimization was employed for both two and three-stage launch vehicle configurations. Three-stage launch vehicle (Case I) was used as a reference of this study. The three-stage launch vehicle had a solid first stage with a predefined M curve thrust profile and upper stages with constant thrust. The two-stage launch vehicles were configured with a throttling capability in steps for both stages. For the two-stage launch vehicle, optimal trajectories were obtained for both 500 km x 500 km circular orbit and 180 km x 36000 km Geostationary Transfer Orbit. The inputs as defined in Eq. (30) are used as target vectors of the problem. Table 4 shows the details of the mission requirement, vehicle definition and launch sites used for the study. The mission constraints used for the solution are as shown in Table 5. Kulasekhara-

patinam was used as the launch site for $\lambda_L = 180^\circ$ mission due to the absence of landmass in the downrange. For the GTO mission, Sriharikota was selected as the launch site since the downrange of Kulasekharapatnam for an eastward launch would confront the landmass. Also, Sriharikota is India's major and only operational spaceport. 585 kgs were considered as the payload, which included 40 kgs for payload structure, 45 kgs for payload electronics and 500 kgs for the satellite. The payload fairing mass considered was 150 kgs. For the circular orbit mission, Gravity Turn was considered up to an altitude of 118 km, the limit of the upper atmosphere. However, for GTO missions, the Gravity Turn trajectory was limited to an altitude of 75 km where the vehicle was still in the atmospheric phase to increase the duration of the optimal trajectory. The angle of attack constraint made sure that the vehicle encountered minimum drag in the upper atmosphere as seen in Fig. 11. I_{sp} was kept constant for a particular stage throughout. If the mission demands, varying I_{sp} could also be included in the program. The thrusts and pitch rates were discretized for two stages for both Case II and Case III as shown in Table 6 and Table 7. The discretization was done in both amplitude and time duration. The simulations were run with a time step of 0.1s.

The values of mutation factor, crossover ratio, dimension and population size used for optimization are shown in Table 8. D and N_p remains constant throughout the generations, whereas F and Cr can be changed in between. Initially, the $F = 0.5$, $Cr = 0.7$ combination was used for the solution for Cases I and II. The convergence was slow. Later, $F = 0.4$ and $Cr = 0.9$ combination was used from generation No. $G = 1.5 \times 10^6$ for Case I and $G = 2.0 \times 10^6$ for Case II. After the change in parameters, the solution converged quickly. The same changed parameters were used for the solution of Case III throughout. The sen-

sitivity of the convergence of the solution w.r.t F & Cr signified the need for an iterative procedure to arrive at the optimal combination. However, the parameters were arrived at in our study through trial and error. These parameters can also be iteratively found out as part of the existing algorithm in future studies. Also, a comprehensive study involving a wide range of parameter variations can be taken up in the future.

All 3 cases of optimization have converged to the required results. The results have converged to the orbit requirement and end/terminal constraint as mentioned in Tables 4 and 5. Stopping criteria were applied based on

the standard deviation of the objective values. Table 10 shows the final results of optimization. The payload mass shown in the table is not inclusive of 150 kgs of the payload fairing, which has been omitted in the ζ_{PL} estimation. The solutions converged as shown in Table 10. Figs. &7–9 show the dynamic parameters of the optimization solutions. The L2 norm of constraints reducing to zero signifies that all candidate solutions meet the feasibility constraints. The L2 norm of fitness is the norm of objective values of the candidate solutions. The reduction in the L2 norm of fitness signifies the reduction in lift-off mass of the vehicle, and when the solution approaches the global optimum,

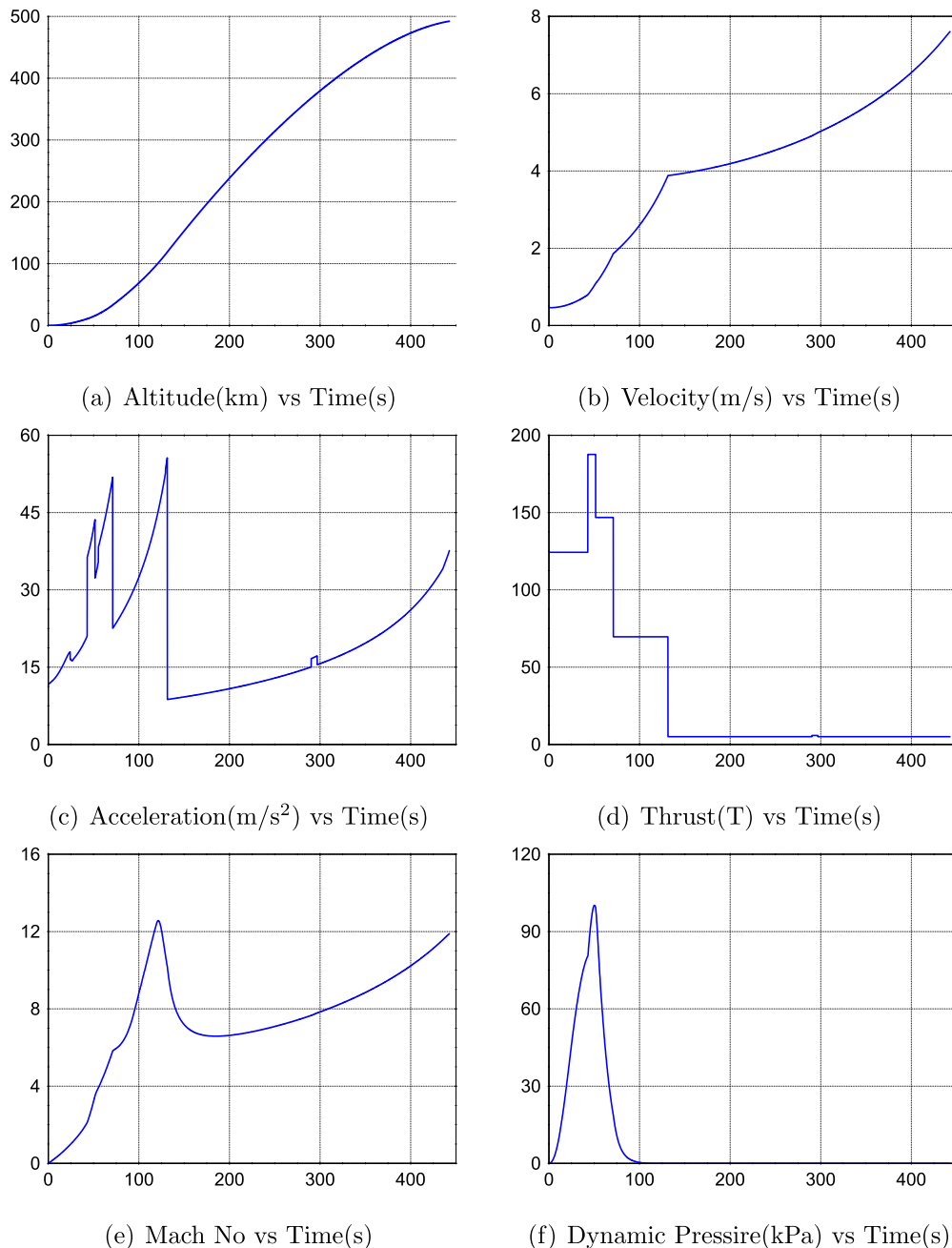


Fig. 10. Expected performance of DE Optimized two-stage launch Vehicle for 500kms×500kms circular orbit.

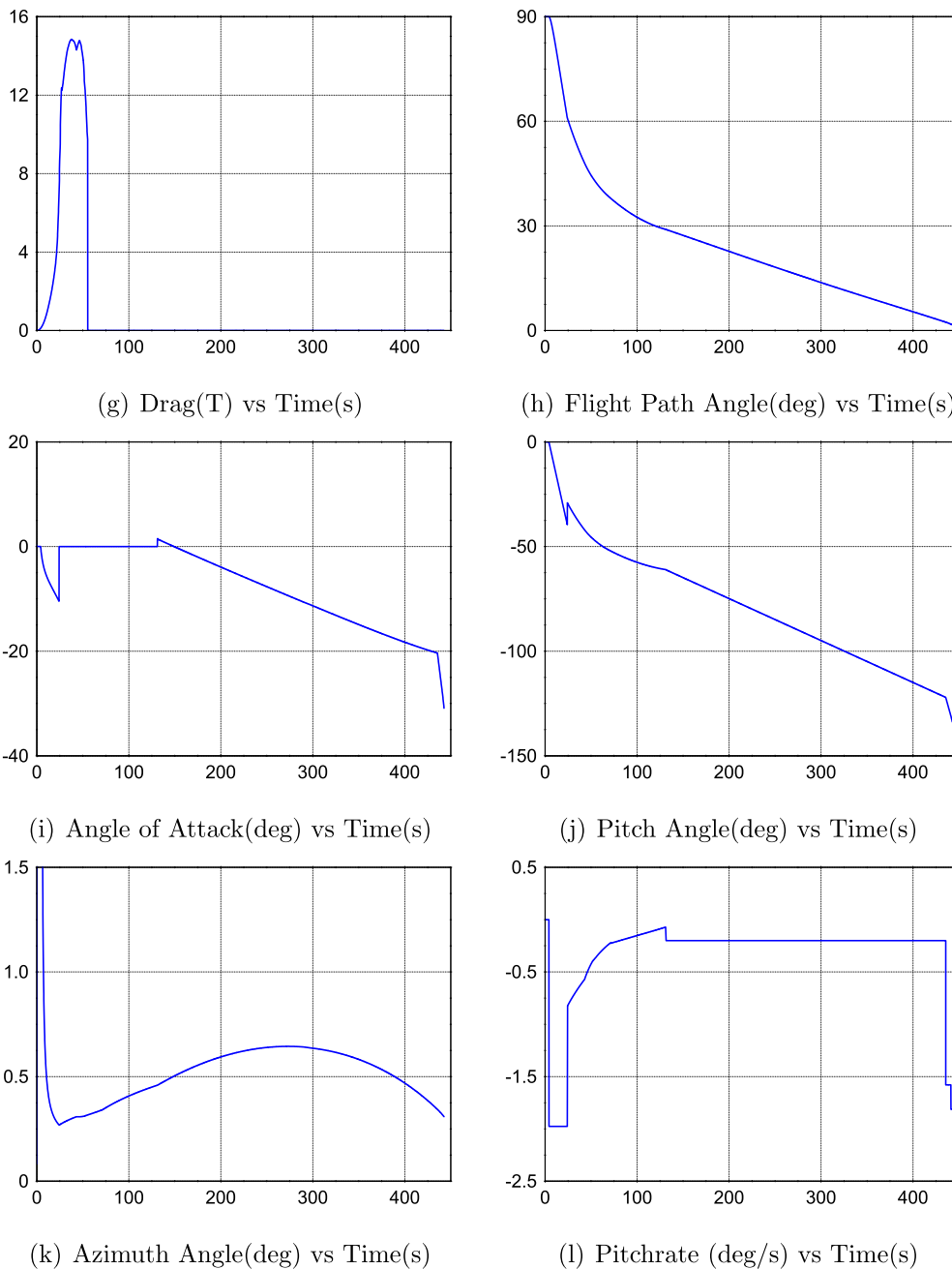


Fig 10. (continued)

the convergence rate increases, as shown in the figures. Once the solution converges to the global optimum, the target vectors and objective values also get converged to the unique optimum due to differential mutation. The solutions were stopped as per the stopping criteria in Section 4.

The expected performance in flight which are the outputs of the optimization for Cases **II** and **III** are shown in Figs. 10 and 11 respectively. In both Case **II** and Case **III**, the optimum result was obtained at the constraint boundary of vehicle acceleration. In case **II**, the convergence lies in the boundary of the dynamic pressure constraint as well, whereas in case **III**, the optimal solution

does not lie on the boundary of the dynamic pressure constraint. It means that relaxed acceleration and dynamic pressure constraints would result in a higher payload fraction for the rocket.

The magnitude of vehicle acceleration is directly dependent on thrust. A higher acceleration constraint means that the vehicle can travel with a higher thrust, and the impulse is delivered within a shorter time, thus reducing the gravity loss. If the steering losses are considered constant in both high and low thrusting cases, the one with higher thrust would have less ideal velocity loss due to gravity. The vehicle velocity depends on the time integral of the vehicle's

acceleration, and dynamic pressure is proportional to the square of the vehicle velocity. The trajectory design moves towards the optima as acceleration increases due to a reduction in burn time. And the result is an increase in dynamic pressure, which is limited by the magnitude of its constraint. For the first stage in both cases, the optimum thrust is observed to decrease in magnitude as the mission progresses in time. As the rocket's mass reduces as time progresses, the thrust needs to be reduced to keep the peak acceleration within its constraint. Hence, the path constraints are the limiting factor for an optimal thrust profile

and trajectory. The relation as mentioned above between the optimal thrust profile and path constraints explains why [Ponssard et al. \(2009\)](#) were able to obtain the optimal trajectory at a constant thrust profile for an optimization that did not employ path constraints.

[Tables 11 and 12](#) highlight the flight sequence for the Case II and Case III missions.

For the first stage in Case II, the thrust varies between 188 T and 70 T, which is a 37% throttling requirement for the stage. For the first stage in Case III, the thrust varies between 305 T and 100 T, which is a 33% throttling

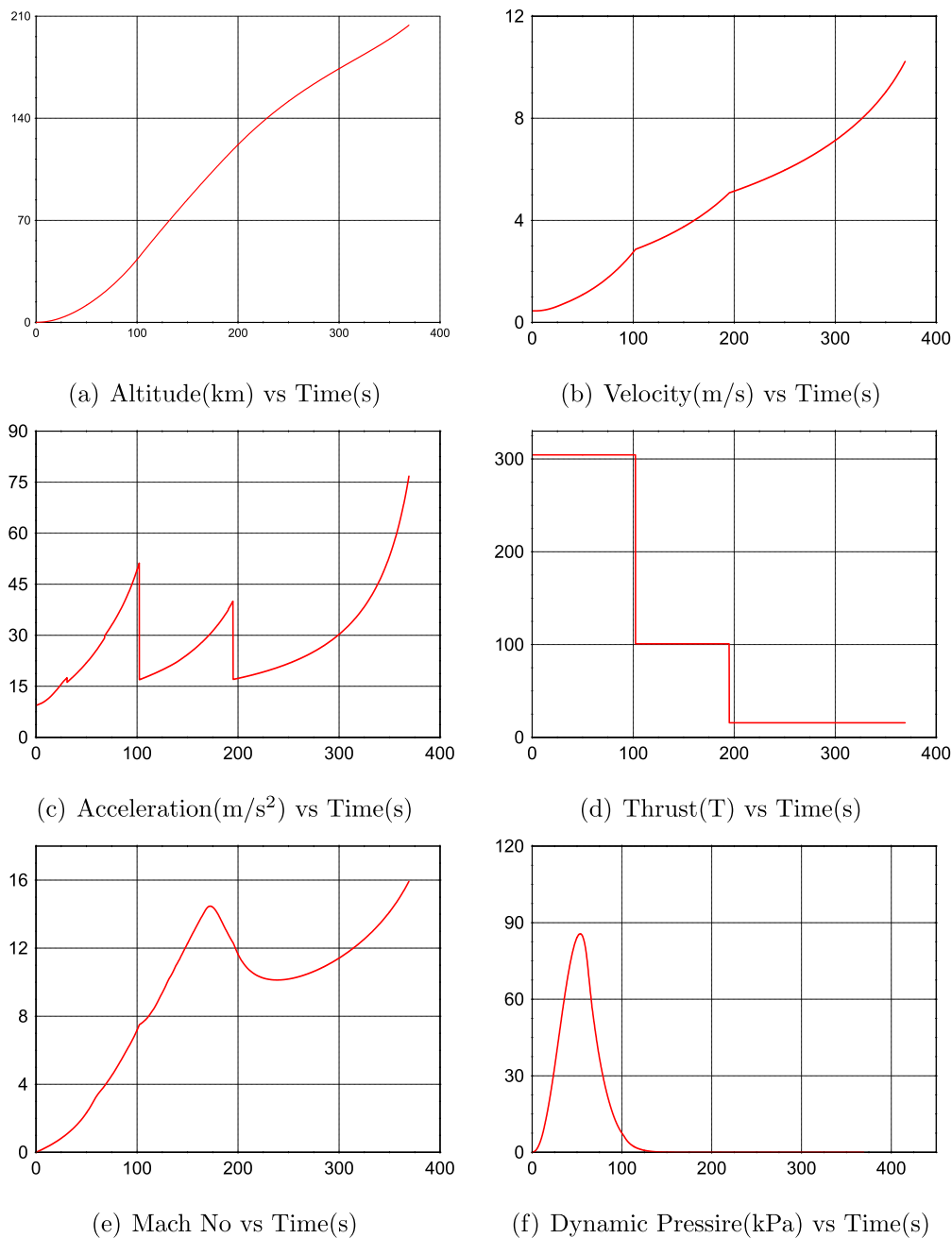


Fig. 11. Expected performance of DE Optimized two-stage launch Vehicle for 180kms×36000kms Geostationary Transfer Orbit.

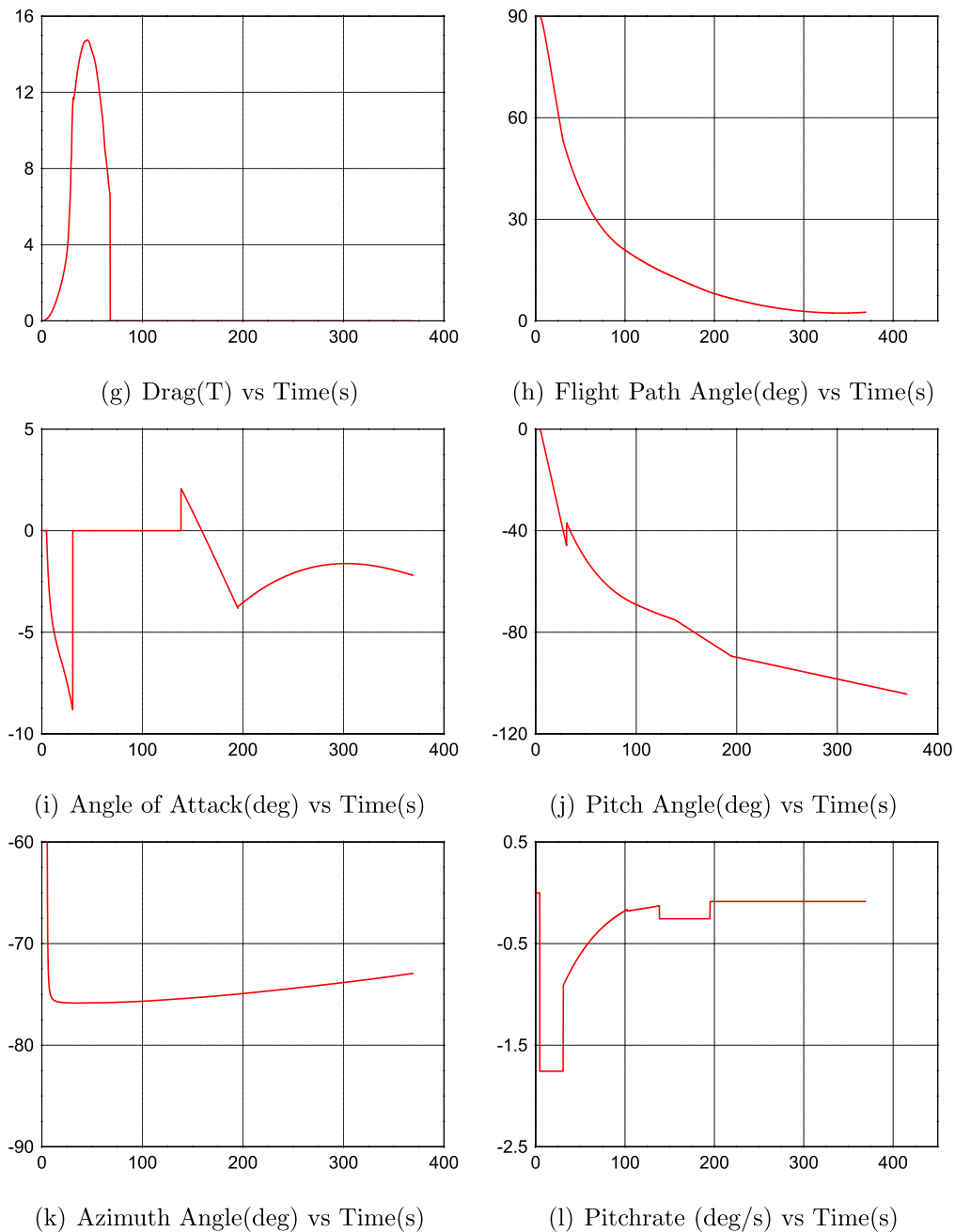


Fig 11. (continued)

requirement for the stage. The second stage thrust remains more or less constant throughout for both the cases at 5.1T for Case **II** and 15.5T for Case **III**.

Lift-off masses of Case **II** and Case **III** have been compared to constant thrust cases with the same vehicle and mission configuration. The results are compared in Table 9. Lift-off mass is reduced by 41.7% for Case **II** and by 16.3% for Case **III**.

Case **II** has an ideal velocity loss of 1536 m/s and Case **III** has 700 m/s. The ideal velocity loss is due to gravity, drag, and steering losses. Even though the thrust axis is kept aligned with the vehicle's longitudinal axis, the loss

in velocity due to steering occurs due to the angle of attack (α).

As shown in Tables 4 and 10 a two-stage throttleable launch vehicle with propellants having 20-40s higher I_{sp} and a structural factor as low as 0.1 can replace a three-stage launch vehicle with solid or earth storable liquid propellants which have I_{sp} in the range of 270–310, which are used in Indian launch vehicles. The advantage of a two-stage launch vehicle over a three-stage launch vehicle is mentioned in Table 13.

In the future, suitable mathematical and computational techniques could automate the manual iterative procedure

Table 11
Case II flight sequence.

Sl. No	Stage	Event	End Time (s)
1	1	Lift off	0.00
2	1	Vertical rise	3.93
3	1	Kickrate maneuver	24.10
4	1	Payload fairing separation	129.31
5	1	Gravity turn	131.00
6	1	Stage 1 burnout and separation	131.30
7	2	Stage 2 burnout and payload injection	442.70

Table 12
Case III flight sequence.

Sl. No	Stage	Event	End Time (s)
1	1	Lift off	0.00
2	1	Vertical rise	4.60
3	1	Kickrate maneuver	30.81
4	1	Gravity turn	138.40
5	1	Payload fairing separation	190.40
6	1	Stage 1 burnout and separation	195.00
7	2	Stage 2 burnout and payload injection	369.20

required to arrive at the optimal F & Cr combination. The optimization process can be stopped at the shallow region of convergence and restarted by retaining the target vector with the best fitness/constraints in the population. The equality constraints shown in Table 5 can be reduced by combining DE and gradient-based optimization methods. As explained in [Chai et al. \(2017a\)](#), the gradient-based optimization can begin considering DE-based optimization's output as its input. Combining these techniques will ensure a quicker and accurate convergence to the global minimum.

6. Conclusion

The study shows that Differential Evolution is a feasible candidate for global optimization problems in the launch vehicle domain. The method has the advantage of no initial guess and rates requirement. Hence, the method is considered one of the best candidates for optimizing a launch vehicle's ascent trajectory and thrust profile simultaneously.

Variable thrusting reduced lift-off mass of the two-stage launch vehicle for 500×500 km orbit by 41.6% and for GTO by 16.3% compared to constant thrusting cases. The thrust variation required in the first stage of the

500 km circular orbit mission and GTO mission points to the requirement of a liquid stage with throttling capability. Pintle type injectors shall be used for the main engines to maintain the required specific impulse and combustion stability margin during throttled operation, in the case of single-engine stages.

Crossover ratio and Mutation factor have a considerable influence on the speed and convergence of the global optimum, which can give an incomplete/non-convergence if not carefully chosen. A two-stage launch vehicle with a structural factor as low as 0.1, 7–15% higher I_{sp} and throttling capability can replace a three-staged launch vehicle for the same payload.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 13
Qualitative comparison of two-stage and three-stage launch vehicles.

Aspect	2 Stages	3 Stages
Separation Events	1	2
Impact points	1	2
Risk of failure	low	high
No. of parts	less	more

the optimization methodology and cluster computing. The guidance from Mr. Hari Kurup, Director, ERT Solutions LLC, Houston, Texas, was essential in implementing parallelism with Python language. Special Thanks to Dr. Sudheesh (Dept. of Physics, IIST) and Dr. Manoj T. Nair (Dept. of Aerospace Engineering, IIST) for the support and facilities provided with High-Performance Computing.

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